

FINS3616 International Business Finance — Exam Formula Sheet

Convention: e = HC/FC (home currency per unit of foreign currency) $e \uparrow$ =home depreciates $e \downarrow$ =home appreciates

NOTATION KEY

e_0 = current spot; e_t = future/expected spot; f_t = forward rate
 r_h, r_f = home/foreign nominal rate; a = real interest rate
 i_h, i_f = home/foreign expected inflation; t = periods
 S_T = spot at expiry; K = strike; c/p = call/put premium
 Bid $\leq$ mid $\leq$ Ask; Dealer: buys at bid, sells at ask

Exchange Rate Basics

Price conversion: $P_{\text{HC}} = P_{\text{FC}} \times e_0$

Percentage change:

$$\% \Delta e = \frac{e_1 - e_0}{e_0} \times 100$$

> 0 : home depreciated. < 0 : home appreciated.

If A depreciates by $x\%$, B appreciates by:

$$\frac{x}{1-x} \times 100\% \quad \text{e.g. } x=0.17 \Rightarrow 20.5\%$$

Reciprocal: $e_{B/A} = 1/e_{A/B}$

Bid–Ask Spreads & Cross Rates

Percentage spread: $\frac{\text{Ask} - \text{Bid}}{\text{Ask}} \times 100$

Cross (no spread): $e_{A/B} = \frac{e_{A/\$}}{e_{B/\$}}$

Cross bid: $\text{Bid}_{B/A} = \frac{\text{Bid}_{B/\$}}{\text{Ask}_{A/\$}}$

Cross ask: $\text{Ask}_{B/A} = \frac{\text{Ask}_{B/\$}}{\text{Bid}_{A/\$}}$

Reciprocal quotes: $\text{Bid}_{A/B} = \frac{1}{\text{Ask}_{B/A}} \quad \text{Ask}_{A/B} = \frac{1}{\text{Bid}_{B/A}}$

Triangular Arbitrage

Implied cross: $e_{A/B}^* = e_{A/\$} / e_{B/\$}$

Compare $e_{A/B}^*$ to quoted $e_{A/B}$: if different $\Rightarrow$ arbitrage.

Buy cheapest route; sell dearest.

Profit = ending amount – starting amount.

Forward Rates

Outright from swap points:

$$F = S_0 + \text{swap pts} \times 0.0001$$

Annualised forward premium (+) / discount (–):

$$\text{Prem/disc} = \frac{F - S_0}{S_0} \times \frac{360}{\text{days}} \times 100$$

+: FC at premium (home expected to depreciate).

–: FC at discount (home expected to appreciate).

Purchasing Power Parity (PPP)

Absolute (Law of One Price): $e_0 = P_h / P_f$

Relative — Exact:

$$e_t = e_0 \left(\frac{1 + i_h}{1 + i_f} \right)^t$$

Higher home inflation $\Rightarrow$ home currency depreciates.

Relative — Approx: $\frac{e_t - e_0}{e_0} \approx i_h - i_f$

Real exchange rate:

$$e_t^{\text{real}} = e_t^{\text{nom}} \left(\frac{1 + i_f}{1 + i_h} \right)^t$$

$e_t^{\text{real}} = e_0$: PPP held. $e_t^{\text{real}} \neq e_0$: PPP failed.

Fisher Effect (FE)

Exact: $(1 + r) = (1 + a)(1 + i)$

Approx: $r \approx a + i$

Real rate: $a = \frac{1 + r}{1 + i} - 1$

Use exact when inflation is large.

Generalised Fisher Effect (GFE)

Real rates equalise internationally: $a_h = a_f$, so:

$$\frac{1 + r_h}{1 + r_f} = \frac{1 + i_h}{1 + i_f}$$

Approx: $r_h - r_f \approx i_h - i_f$

Nominal rate diff $\approx$ inflation diff. *GFE bridges PPP and IFE.*

International Fisher Effect (IFE)

Exact: $e_t = e_0 \left(\frac{1 + r_h}{1 + r_f} \right)^t$

Approx: $\frac{e_t - e_0}{e_0} \approx r_h - r_f$

Home $\uparrow$ nominal rate $\Rightarrow$ home currency depreciates.

Risky (uncovered): actual spot may differ.

Solving for unknown rate: $r_h = \left(\frac{e_t}{e_0} \right)^{1/t} (1 + r_f) - 1$

Interest Rate Parity (IRP)

No-arb condition (exact):

$$f_t = e_0 \left(\frac{1 + r_h}{1 + r_f} \right)^t$$

Approx: $\frac{f_t - e_0}{e_0} \approx r_h - r_f$

IRP = IFE formula but LHS is *forward* rate (riskless, locked in).

Enforced by covered interest arbitrage (CIA).

UFR: IRP + IFE both hold $\Rightarrow f_t = e_t$

Parity Conditions Summary

PPP: $e_t = e_0 \left(\frac{1 + i_h}{1 + i_f} \right)^t$ Approx: $\Delta\%e \approx i_h - i_f$

Fisher: $(1 + r) = (1 + a)(1 + i)$ Approx: $r \approx a + i$

GFE: $\frac{1 + r_h}{1 + r_f} = \frac{1 + i_h}{1 + i_f}$ Approx: $r_h - r_f \approx i_h - i_f$

IFE: $e_t = e_0 \left(\frac{1 + r_h}{1 + r_f} \right)^t$ Approx: $\Delta\%e \approx r_h - r_f$

IRP: $f_t = e_0 \left(\frac{1 + r_h}{1 + r_f} \right)^t$ Approx: $\Delta\%f \approx r_h - r_f$

UFR: $f_t = e_t$ (if IRP and IFE both hold)

Covered Interest Arbitrage (CIA)

Step 1: Compute $f_t^* = e_0(1 + r_h)/(1 + r_f)$.

If $f_t > f_t^*$: borrow HC $\rightarrow$ buy FC $\rightarrow$ invest at $r_f \rightarrow$ sell FC forward.

If $f_t < f_t^*$: borrow FC $\rightarrow$ buy HC $\rightarrow$ invest at $r_h \rightarrow$ sell HC forward.

Hedged FC return (per unit HC):

$$R = \frac{1}{e_0} (1 + r_f) \cdot f_t - (1 + r_h)$$

$R > 0$: arb profit. $R < 0$: no arb.

Transaction cost δ per leg kills arb if $R < 2\delta$.

Currency Futures

Daily P/L: (today’s price – prev. settlement) $\times$ contr. size

Total P/L: $(P_{\text{final}} - P_{\text{entry}}) \times \text{contract size}$

Contracts: FC exposure $\div$ contract size

Payable FC $\Rightarrow$ buy (long); Receivable FC $\Rightarrow$ sell (short).

Margin call $\rightarrow$ top up to **initial** margin (not maintenance).

Futures–forward arb profit: $|P_{\text{fut}} - P_{\text{fwd}}| \times \text{contract size}$

Currency Options

Long call: $\pi = \max(S_T - K, 0) - c$ B/E: $S_T = K + c$

Short call: $\pi = c - \max(S_T - K, 0)$ max gain = c

Long put: $\pi = \max(K - S_T, 0) - p$ B/E: $S_T = K - p$

Short put: $\pi = p - \max(K - S_T, 0)$ max gain = p

ITM: call if $S_T > K$; put if $S_T < K$.

OTM: call if $S_T < K$; put if $S_T > K$.

Total profit = per-unit profit $\times$ contract size.

Known exposure $\Rightarrow$ futures. **Contingent** $\Rightarrow$ option.

Swaps

Total gain from swap: $\Delta_{\text{fixed}} - \Delta_{\text{floating}} > 0$

Each party borrows where it has comparative advantage.

IR swap: same currency; **no** principal exchange.

Currency swap: diff. currencies; principal **is** exchanged.

Trap: yen 6% vs. USD 10% is not “free 4%” — lower yen rate reflects expected yen appreciation (IFE).

EXAM TRAPS — Read before calculating

- **Convention:** e = HC/FC; $e \uparrow$ =home depreciates; $e \downarrow$ =home appreciates.
- **Asymmetric:** A depreciates $x\% \Rightarrow$ B appreciates $x/(1-x) \times 100\%$ (NOT $x\%$).
- **Dealer:** buys at *bid*, sells at *ask*; customer buys FC at *ask*, sells FC at *bid*.
- **Reciprocal bid/ask:** $\text{Bid}_{A/B} = 1/\text{Ask}_{B/A}$; $\text{Ask}_{A/B} = 1/\text{Bid}_{B/A}$.
- **Cross rates:** cross *bid* = bid/**ask**; cross *ask* = ask/**bid**.
- **Period matching:** use periodic rate, not annual (e.g. 3% for 6-month contract if annual = 6%).

- **Decimals:** convert % to decimals before substituting (6% $\rightarrow$ 0.06).
- **Exact vs. approx:** use exact unless question says “approximately”; critical at large rates.
- **360-day rule:** forward premium/discount formula uses 360 days.
- **Margin call:** top up to *initial* margin, not maintenance margin.
- **IFE vs. IRP:** same RHS — e_t (expected spot, risky/uncovered) vs. f_t (forward, locked in/covered).
- **Currency swap trap:** comparing nominal rates in different currencies is misleading without adjusting for expected exchange rate changes (IFE).